

Confidence limit for Proportions

By Central Limit Theorem, we have

$$Z = \frac{t - E(t)}{SE(t)} \sim N(0,1)$$

Substitute $t=p$ where p is the proportion.

$$\text{then } Z = \frac{p - E(p)}{SE(p)} \sim N(0,1)$$

We have $E(p) = P$ & $SE(p) = \sqrt{PQ/n}$
where $Q = 1 - P$

$$\text{ie } Z = \frac{p - P}{\sqrt{PQ/n}} \sim N(0,1)$$

$100(1-\alpha)\%$ is ^{the} given ~~by~~ confidence limit
for p

Confidence Interval of proportion is

$$p \pm Z_{\alpha/2} \sqrt{PQ/n}$$

$$95\% \text{ CI is } p \pm 1.96 \sqrt{PQ/n}$$

$$99\% \text{ CI is } p \pm 2.58 \sqrt{PQ/n}$$

Notes :- Confidence interval for proportion

→ If no estimate of population proportion 'p' is available, in such situations $p=0.5$ & $q=0.5$.

$$SE = \sqrt{\frac{pq}{n}}$$

$$= \sqrt{\frac{0.5 \times 0.5}{n}}$$

$$= \frac{0.5}{\sqrt{n}}$$

→ In case of confidence limit for proportion, if the population size N is given, then the standard error is given by,

$$SE(p) = \sqrt{\frac{(N-n)pq}{N(n-1)}}$$

Or in special cases,

$$SE(p) = \sqrt{\frac{(N-n)pq}{n(N-1)}}$$

where N is the population size,

n is the sample size,

p is the proportion

$$q = 1 - p.$$

The confidence limits can be found out by,

$$p \pm Z_{\alpha/2} \sqrt{\frac{(N-n)pq}{N(n-1)}}$$

and this is applicable for simple random sampling without replacement.

Q: -1 A random sample of 100 items taken from a batch or articles containing 5 defective items

a) Set 96% confidence limits for the proportion of defective items in the batch.

b) If the batch contains 2669 items setup 95% confidence limits for the proportion of defective items.

Given $n = 100$,

$$p = \text{Proportion of defective} = \frac{5}{100} = 0.05$$

$$q = 1 - p = 1 - 0.05 = 0.95$$

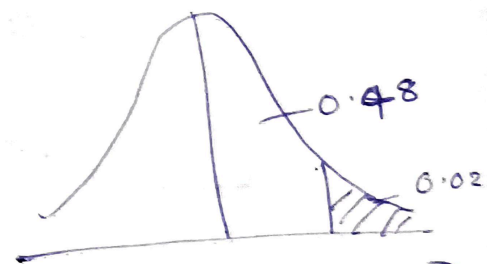
a) Given $100(1 - \alpha)\% = 96\%$

$$1 - \alpha = 0.96$$

$$\alpha = 1 - 0.96 = 0.04$$

$$\alpha/2 = 0.04/2 = \underline{\underline{0.02}}$$

$$\text{ie, } P(Z > Z_{\alpha/2}) = 0.02.$$



$$\text{ie } Z_{\alpha/2} = 2.05 \leftarrow \text{[From std normal tables]}$$

C.I on proportion is

$$p \pm Z_{\alpha/2} \cdot \sqrt{\frac{pq}{n}}$$

$$p - Z_{\alpha/2} \cdot \sqrt{\frac{pq}{n}} \leq p \leq p + Z_{\alpha/2} \cdot \sqrt{\frac{pq}{n}}$$

$$0.05 - 2.05 \sqrt{\frac{0.05 \times 0.95}{10}} \leq p \leq 0.05 + 2.05 \sqrt{\frac{0.05 \times 0.95}{10}}$$

$$\underline{\underline{0.0054 \leq p \leq 0.0946}}$$

b) Here $N = 2669$, $n = 100$, $P = \frac{5}{100} = 0.05$, $q = 0.95$

$$100(1-\alpha)\% = 95\%$$

$$\alpha = 0.05$$

$$\alpha/2 = 0.025$$

$$Z_{\alpha/2} = 1.96$$

Confidence Interval on proportion is given by

$$P \pm Z_{\alpha/2} \cdot \sqrt{\frac{(N-n)Pq}{N(n-1)}}$$

$$\text{i.e. } 0.05 \pm 1.96 \sqrt{\frac{(2669-100)0.05 \times 0.95}{2669(100-1)}}$$

$$= 0.05 \pm 1.96(0.0214)$$

$$= 0.05 \pm 0.0419$$

$$\text{i.e., } \underline{\underline{0.0081 \leq P \leq 0.0919}}$$

Q2. A random sample of 700 units from a large consignment showed that 200 were damaged. Find 95% & 99% confidence limits for the proportion of damaged units.

Given $n = 700$ $p = \frac{200}{700} = \frac{2}{7} = 0.286$

$$q = 1 - p = \frac{5}{7} = 0.714$$

To find 95% CI

$$\alpha = 0.05$$

$$\alpha/2 = 0.025$$

$$Z_{\alpha/2} = 1.96$$

CI on proportion is $p \pm \sqrt{\frac{pq}{n}} \cdot Z_{\alpha/2}$

$$\text{ie } 0.286 \pm 1.96 \sqrt{\frac{0.286 \times 0.714}{700}}$$

$$\Rightarrow 0.286 \pm 0.0332$$

$$\Rightarrow \underline{0.2468 \leq p \leq 0.3132}$$

To find 99% CI, $\alpha = 0.01$, $\alpha/2 = 0.005$

$$\therefore Z_{\alpha/2} = 2.58$$

CI on proportion is

$$0.286 \pm 2.58 \sqrt{\frac{0.286 \times 0.714}{700}}$$

$$\text{ie, } \underline{0.2363 \leq p \leq 0.3237}$$

